A natural question that arises in the context of chaotic systems is whether such a trajectory average in fact makes sense. Considering any conservative system, we know that the energy itself is a conserved quantity along trajectories, and its average is therefore a function of the initial condition. On the other hand, imagine trajectories emanating from different points on a single energy level set Σ_{E}, can we necessarily conclude that all such trajectories will yield the same average value for any observable? This question is obviously important as the initial conditions of a molecular system are typically generated through approximations and estimations—it would be disturbing if the choice of these (at uniform energy) were to greatly impact the computation of an average.

One situation in which we would expect different initial conditions to yield different trajectory averages is if the energy surface has several connected components.

Example 5.3 Consider the *double well potential*

$$\phi(q) = (1 - q^2)^2.$$

The energy $H(q, p) = p^2/2 + \phi(q)$ will tend to $+\infty$ as $q \to \infty$ or $p \to \infty$, but is obviously bounded below (by zero). The graph of the energy, along with some level curves, is shown in Fig. 5.2. From the graph it is clear that the energy surface $H(q, p) = \mathrm{E}$ has more than one component if $0 < \mathrm{E} < 1$. If an initial condition is taken from one of the two basins, say with energy E_0, then the trajectory through that point will remain forever in the one basin, thus the averages of many quantities, if computed using the trajectory values, would not be a good approximation of the overall average at constant energy E_0.

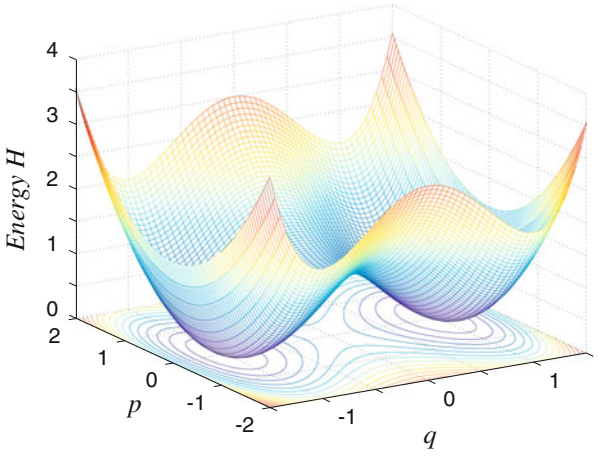


Fig. 5.2 Energy function for the double well. The energy surface corresponding to a value of energy $\mathrm{E} \in (0, 1)$ is not connected